

Companion to: *A Band-Mask Robustness Diagnostic for LDA on HSI*

2026

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B.2.2 Generative model

The standard LDA generative process [2] applies verbatim to the band-frequency corpus. For K topics and a vocabulary of size V , with hyperparameters $\alpha > 0, \eta > 0$:

$$\phi_k \sim \text{Dirichlet}(\eta \mathbf{1}_V), \quad k = 1, \dots, K, \quad (3)$$

$$\theta_d \sim \text{Dirichlet}(\alpha \mathbf{1}_K), \quad d = 1, \dots, D, \quad (4)$$

$$z_{d,n} \sim \text{Categorical}(\theta_d), \quad n = 1, \dots, N_d, \quad (5)$$

$$w_{d,n} \sim \text{Categorical}(\phi_{z_{d,n}}). \quad (6)$$

The joint posterior $p(\theta, \phi, z \mid w, \alpha, \eta)$ is intractable; we approximate it with the online variational Bayes inference of Hoffman, Blei, and Bach [3].

B.2.3 Online VB lower bound

Let $q(\theta_d, z_d) = q(\theta_d \mid \gamma_d) \prod_n q(z_{d,n} \mid \lambda_{d,n})$ be the document-local variational distribution and $q(\phi_k) = q(\phi_k \mid \beta_k)$ the topic-word distribution. The evidence lower bound (ELBO) on document d is

$$\begin{aligned} \mathcal{L}_d(\gamma_d, \lambda_d, \beta) &= \mathbb{E}_q[\log p(w_d, z_d, \theta_d, \phi)] \\ &\quad - \mathbb{E}_q[\log q(z_d, \theta_d, \phi)], \end{aligned} \quad (7)$$

which after expansion and use of the digamma identity $\mathbb{E}_{\text{Dirichlet}}[\log \pi_k] = \psi(\alpha_k) - \psi(\sum_{k'} \alpha_{k'})$ takes the form

$$\begin{aligned} \mathcal{L}_d &= \sum_{n=1}^{N_d} \sum_{k=1}^K \lambda_{d,n,k} [\psi(\gamma_{d,k}) - \psi(\sum_{k'} \gamma_{d,k'}) \\ &\quad + \psi(\beta_{k,w_{d,n}}) - \psi(\sum_v \beta_{k,v})] \\ &\quad - \sum_{n,k} \lambda_{d,n,k} \log \lambda_{d,n,k} + C(\alpha, \eta, \gamma_d, \beta). \end{aligned} \quad (8)$$

Maximisation w.r.t. λ yields

$$\lambda_{d,n,k}^* \propto \exp[\psi(\gamma_{d,k}) + \psi(\beta_{k,w_{d,n}}) - \psi(\sum_v \beta_{k,v})]. \quad (9)$$

Maximisation w.r.t. γ yields

$$\gamma_{d,k}^* = \alpha + \sum_n \lambda_{d,n,k}^*. \quad (10)$$

The global β is updated via natural-gradient ascent on the stochastic mini-batch ELBO with the learning-rate schedule $\rho_t = (\tau_0 + t)^{-\kappa}$, $\tau_0 = 10, \kappa = 0.7$, mini-batch size 128, $T_{\max} = 60$ passes over the corpus.

B.2.4 Posterior summaries used in the diagnostic

After convergence we extract:

$$\phi_{k,b} = \frac{\beta_{k,b}}{\sum_v \beta_{k,v}}, \quad \text{topic-word distributions,} \quad (11)$$

$$\theta_{d,k} = \frac{\gamma_{d,k}}{\sum_{k'} \gamma_{d,k'}}, \quad \text{per-document topic posterior,} \quad (12)$$

$$z_d = \arg \max_k \theta_{d,k}, \quad \text{dominant topic per document.} \quad (13)$$

C.3 Band-mask sweep

C.3.1 Mask specification

A band mask is a function $m : \{0, \dots, B-1\} \rightarrow \{0, 1\}$ with $m(b) = 1$ if band b is retained. The masked document-term matrix restricts the count vector accordingly: $\mathbf{c}_d^{(m)} = (c_{d,b})_{b:m(b)=1}$. The masked vocabulary size is $V^{(m)} = |\{b : m(b) = 1\}|$.

The four canonical masks of the diagnostic are:

$$m_{\text{vnir}}(b) = \mathbf{1}[\lambda_b \in [400, 1100] \text{ nm}], \quad (14)$$

$$m_{\text{swir}}(b) = \mathbf{1}[\lambda_b \in [1100, 2500] \text{ nm}], \quad (15)$$

$$m_{\text{nw}}(b) = 1 - \mathbf{1}[\lambda_b \in W], \quad (16)$$

$$m_{\text{f50}}(b) = \mathbf{1}[b \in \text{TopK}_{50}(F)], \quad (17)$$

where $W = [1350, 1430] \cup [1800, 1950] \cup [2480, 2500] \text{ nm}$ and F_b is the per-band Fisher discriminant ratio

$$F_b = \frac{\sum_{\ell} n_{\ell} (\mu_{b,\ell} - \mu_b)^2}{\sum_{\ell} n_{\ell} \sigma_{b,\ell}^2} \quad (18)$$

with $\mu_{b,\ell}, \sigma_{b,\ell}$ the per-class mean and standard deviation of reflectance at band b for class ℓ , computed on the *unmasked* spectrum.

C.3.2 Masked refit

For each m we refit the LDA model with the same $K, \alpha, \eta, T_{\text{max}}$, seed as the canonical fit but using the restricted document-term matrix. This yields $\phi^{(m)}, \theta^{(m)}$, and a masked dominant-topic map $z_d^{(m)}$.

D.4 Paired adjusted Rand index

Definition D.4.1 (Adjusted Rand index). Let $U = \{U_1, \dots, U_K\}$ and $V = \{V_1, \dots, V_{K'}\}$ be two partitions of the same set $\mathcal{S}$. Let $n_{kj} = |U_k \cap V_j|$, $n_k = |U_k|$, $n_j = |V_j|$, $n = |\mathcal{S}|$. The Rand index is $R = (a + b) / \binom{n}{2}$ where a counts pairs in the same cluster under both U and V and b counts pairs in different clusters under both. The adjusted Rand index corrects for chance via the standard form

$$\text{ARI}(U, V) = \frac{\sum_{k,j} n_{kj} \binom{n_{kj}}{2} - [\sum_k \binom{n_k}{2} \sum_j \binom{n_j}{2}] / \binom{n}{2}}{\frac{1}{2} [\sum_k \binom{n_k}{2} + \sum_j \binom{n_j}{2}] - [\sum_k \binom{n_k}{2} \sum_j \binom{n_j}{2}] / \binom{n}{2}}.$$

The paired-ARI in the diagnostic is

$$\text{ARI}_{\text{paired}}(m) = \text{ARI}(\{z_d\}_{d \in \mathcal{S}}, \{z_d^{(m)}\}_{d \in \mathcal{S}}), \quad (19)$$

with $\mathcal{S}$ the set of pixels labelled in both the canonical and masked dominant-topic maps (sentinel 255 excluded).

E.5 Hungarian topic-id alignment

ARI is invariant under topic-id permutations of V , so it summarises how much agreement exists but not which masked topic corresponds to which canonical topic. We recover the correspondence by solving the maximum-weight bipartite matching on the confusion matrix.

Definition E.5.1 (Confusion matrix). $C \in \mathbb{Z}_{\geq 0}^{K \times K^{(m)}}$ with $C_{k,j} = |\{d \in \mathcal{S} : z_d = k \wedge z_d^{(m)} = j\}|$.

Definition E.5.2 (Hungarian permutation). $\sigma^* : \{0, \dots, K-1\} \rightarrow \{0, \dots, K^{(m)}-1\}$ is the solution of the linear-assignment problem

$$\sigma^* = \arg \max_{\sigma} \sum_{k=0}^{K-1} C_{k, \sigma(k)}, \quad (20)$$

computed in $O(K^3)$ time by the Hungarian algorithm [4], as implemented in `scipy.optimize.linear_sum_assignment` [5, 6].

The swap rate under alignment is

$$\text{swap_rate}(m) = \frac{1}{|\mathcal{S}|} \sum_{d \in \mathcal{S}} \mathbf{1}[z_d^{(m)} \neq \sigma^*(z_d)], \quad (21)$$

the fraction of pixels whose dominant topic still changes after the masked topic ids have been relabelled to maximise diagonal mass.

Proposition E.5.3 (Relationship between paired ARI and swap rate). *Under the canonical $K = K^{(m)}$ assumption, a paired ARI of 1 corresponds to swap rate 0, and vice versa. For $\text{ARI}_{\text{paired}}(m) < 1$ the swap rate strictly bounds the disagreement in pixel labels under the best permutation; ARI in addition adjusts for the chance baseline given the marginal cluster sizes.*

F.6 Diagnostic pseudocode

Algorithm 1 Band-mask robustness diagnostic

Require: Pixel spectra $\mathbf{X} \in \mathbb{R}^{D \times B}$, ground-truth labels $\mathbf{y} \in \mathbb{Z}^D$, wavelengths $\boldsymbol{\lambda} \in \mathbb{R}^B$.

- 1: $\mathbf{C} \leftarrow \text{Tokenise}(\mathbf{X}; Q = 12)$ $\triangleright D \times B$ counts
- 2: $\theta, \phi \leftarrow \text{OnlineVB-LDA}(\mathbf{C}; K, \alpha, \eta, T_{\max}, \text{seed})$
- 3: $z_d \leftarrow \arg \max_k \theta_{d,k}$ for $d = 1, \dots, D$ $\triangleright$ canonical map
- 4: Compute Fisher rank F via Eq. 18.
- 5: **for** $m \in \{\text{vnir}, \text{swir}, \text{nw}, \text{f50}\}$ **do**
- 6: $\mathbf{C}^{(m)} \leftarrow$ retain columns of $\mathbf{C}$ satisfying $m(b) = 1$
- 7: **if** $V^{(m)} < 8$ **then**
- 8: **continue** $\triangleright$ auto-skip Pavia U $\times$ SWIR
- 9: **end if**
- 10: $\theta^{(m)}, \phi^{(m)} \leftarrow \text{OnlineVB-LDA}(\mathbf{C}^{(m)}; K, \alpha, \eta, T_{\max}, \text{seed})$
- 11: $z_d^{(m)} \leftarrow \arg \max_k \theta_{d,k}^{(m)}$
- 12: $\mathcal{S} \leftarrow \{d : z_d \neq 255 \wedge z_d^{(m)} \neq 255\}$
- 13: $\text{ARI}_p(m) \leftarrow \text{ARI}(\{z_d\}_{\mathcal{S}}, \{z_d^{(m)}\}_{\mathcal{S}})$
- 14: Build $C \in \mathbb{Z}^{K \times K}$ from $\mathcal{S}$.
- 15: $\sigma^*(m) \leftarrow \text{LINEARSUMASSIGNMENT}(-C)$
- 16: $\text{swap}(m) \leftarrow |\mathcal{S}|^{-1} \sum_{\mathcal{S}} \mathbf{1}[z_d^{(m)} \neq \sigma^*(z_d)]$
- 17: **end for**
- 18: **return** $\{\text{ARI}_p(m), \sigma^*(m), \text{swap}(m)\}_m$

G.7 Numerical complexity

Per (scene, mask) tuple: one OnlineVB-LDA fit at $O(T_{\max} \cdot D \cdot V^{(m)} \cdot K)$, one $K \times K^{(m)}$ confusion-matrix construction at $O(|\mathcal{S}|)$, one Hungarian assignment at $O(K^3)$. On the largest labelled scene (Salinas, $D = 3520, K = 12, V = 204, T_{\max} = 60$) the canonical-vs-masked diagnostic completes in ~ 12 seconds on a single CPU core.

References

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